

Predicting California Wildfires using Environmental Data: A Comparative Study of Ensemble and Neural Network Models

Isabelle Li Shuang Yau
School of Computing and Artificial
Intelligence
Sunway University
Selangor, Malaysia
23021165@imail.sunway.edu.my

Peggy En Ki Liow
School of Computing and Artificial
Intelligence
Sunway University
Selangor, Malaysia
23020852@imail.sunway.edu.my

Yan Yi Leow
School of Computing and Artificial
Intelligence
Sunway University
Selangor, Malaysia
23020324@imail.sunway.edu.my

Abstract—Over the last few years, wildfires in California have had a significant negative effect on ecosystems, infrastructures, and human life. Since the frequency of wildfire occurrences has been increasing continuously over the years, forecasting wildfires has become more important to provide early warning and mitigation measures. Therefore, this paper combines various base learners and hyperparameter-optimized neural networks on the environmental data of 262 CIMIS weather stations in the years 2020 to 2025. In this case, the preprocessing stage includes outlier removal, normalization, and balancing of classes to ensure the reliability of the model. There are also various ways to evaluate the models, including misclassification rate, accuracy, precision, recall, F1-score, and ROC. The output of this paper shows that neural networks have better performance on recall and total predictive accuracy, while ensemble models provide better precision compared to each other. This paper also shows the significant factors that may affect the environment, including solar radiation, soil temperature, and air temperature. Therefore, this paper provides practical information to wildfire management and evidence-based decision support.

Keywords—Wildfire Prediction, Ensemble Learning, Gradient Boosting, Decision Tree, Regression Models, Neural Networks, Environmental Risk Prediction, Machine Learning in Environmental Science

I. INTRODUCTION

A. Background

In recent years, climate change has been contributing to rising weather temperatures and reduced summer rainfall due to human actions. This increases the likelihood of wildfire occurrence, where wildfire are uncontrolled fires that often occur in wildland vegetation and may pose serious threats to human life and ecosystems [1][2]. The statistics by the California Department of Forestry and Fire Protection reported that in 2025, there were 8019 wildland fires, which is slightly higher than that of 2024 (7794 fire) and the 5-year average (7517 fires). This indicates that the frequency of wildfire incidents is expected to increase in the next few years [3]. It was well known that the most destructive wildfire in California history is the 2018 Camp Fire in Northern California's Butte County, where 153336 acres were burned, 18804 structures

were destroyed and 85 people died [4]. Consequently, numerous studies have focused on wildfire risk prediction to enable early detection and support mitigation strategies to handle this problem.

B. Problem Statement

Various studies have focused on wildlife prediction in California using different types of machine learning models. One of the studies is by Malik et al., who proposed two machine learning architectures, which are ensemble model using stacking and AdaBoost, as well as a combined model, to perform region-based wildfire risk analysis and prediction. This study addresses the limitations of existing studies, where models often lack the processing capabilities needed for real-time wildfire decision making through providing hourly or biweekly risk predictions. Besides, the study incorporated parameters such as powerline to address the limitation of existing models that rely on a narrow range of data parameters, which may fail to provide a comprehensive perspective on wildfire risk [5].

While Malik et al. demonstrated the effectiveness of ensemble-based architectures for real-time wildfire risk prediction, a significant gap remains in determining whether neural networks or ensemble models perform better when processing environmental data. Hence, this study addresses the research gap by conducting a comparative analysis between ensemble model (decision tree, gradient boosting and regression) and neural network model. Addressing this gap is significant as accurate wildfire prediction models can support early warning systems and enhance wildfire management strategies. Moreover, most of the previous studies have focused on a narrow geographical area, such as the localized analysis of California's Monticello and Winters regions. This study addresses the gap by performing analysis on environmental data from 262 CIMIS weather stations across California, in order to develop a generalized framework which ensures reliable wildfire prediction at state-wide level.

C. Research Questions

In order to address the identified gaps in the literature, this study seeks to answer the following research questions:

1. How does the predictive performance of ensemble models compare with neural network models in predicting wildfire risk using environmental data from CIMIS weather stations?
2. Which specific environmental conditions contribute most to wildfire risk prediction during the 2020-2025 period?
3. Which machine learning model provides the best combination of prediction performance and interpretability for early wildfire warning systems, given California's evolving wildfire patterns?

D. Research Aim and Objectives

The aim of this study is to compare the predictive performance of ensemble models and neural network models for wildfire risk prediction using environmental data from CIMIS weather stations across California, to identify the best model for early warning systems. The objectives of this study include:

1. To implement and evaluate ensemble models and neural network models for predicting wildfire risk.
2. To determine the most significant environmental predictors of wildfire risk in California.
3. To recommend wildfire mitigation strategies based on the predictive results from ensemble and neural network models.

II. BACKGROUND / LITERATURE REVIEW

A. Environmental Sustainability Problems and the Role of Data Mining

The issue of global sustainability has become a significant concern due to the increase in climate change, environmental pollution, and the continuous usage of unsustainable resources. People find out that climate hazards such as wildfires, floods, and extreme temperatures have led to serious negative effects on the ecosystem, infrastructure, and human health [6]. In this case, people are trying to control it by using data prediction, but these environmental issues are hard to forecast and control due to the complex interaction between meteorological conditions, land cover features, and human activities [7].

On the other hand, there has been rapid development in large-scale environmental datasets over the past few years due to the improvement of remote sensing technologies, environmental monitoring systems, and data collection platforms. The provision of spatial and temporal details on the environment that may help to predict and control environmental issues has been provided by geospatial data, satellite imagery, and meteorology [8]. However, the volume and complexity of these datasets have been increasing

continuously and often exceeding the analysis capacity of traditional statistical methods.

Therefore, machine learning and data mining methods are becoming more important to overcome the issues in environmental datasets. These methods are able to figure out the hidden patterns, capture non-linear relationships, and deal with high-dimensional environmental data better compared to traditional models [9]. The evidence-based practices are also useful for risk forecasting, early warning models, and long-term decision-making in different environmental applications, including wildfire prediction, pollution control, and ecological risk evaluation [10].

Thus, data mining can be used to transform complex environmental data into practical insights and provide important evidence for evidence-based environmental sustainability management.

B. Traditional Predictive Models

Traditional predictive methods have been popular in environmental datasets due to their simplicity, interpretability, and less computation. In this case, the regression models are one of the most often used models to investigate the relationship between the environmental variables and the target outcomes. For example, the occurrence of hazards and the level of pollution. These models are often used due to their clear mathematicalization, which may help decision makers to understand how the related variables may influence the predicted outcomes. To test a regression model is suitable in specific cases, the relationship between variables should be approximately linear [6].

Moreover, a decision tree is another popular traditional predictive model in environmental forecasting. Unlike regression models, decision trees are allowed to split the data into homogeneous subsets to structure a nonlinear relationship. This format usually enables decision trees to store threshold-based environmental patterns and provides an instinctive explanation. However, single decision tree models are more likely to be affected by noise and also sensitive to changes in training data. These limitations may lead to overfitting and poor generalisation performance while dealing with complex environmental data [11].

Overall, regression and decision tree models may have advantages in terms of understandability and computational efficiency. However, their prediction ability on high-dimensional, non-linear, and large-scale environmental data is inadequate in most cases.

C. Ensemble Learning Techniques

Ensemble learning has become an effective method that enhances the predictive ability in multifaceted environmental data. It addresses the limitations of individual models, such as overfitting and poor extrapolation. Ensemble methods that are often used now are Bagging, Random Forests, and Gradient Boosting, which structure models based on multiple base learners to provide more accurate and stable predictive outcomes. For example, bagging is used to address the overfitting issues in single decision trees by training bootstrap

samples on multiple decision trees and taking the average value [12]. Meanwhile, random forests are the expandable version of this principle. It chooses features randomly while splitting the samples to enhance the diversity and predictive performance of the model and also makes itself more suitable for environmental risk classification issues [13].

Furthermore, gradient boosting algorithms are sequential learners that every successive model may correct the mistakes that previous models made, including XGBoost. It is proficient in capturing complex non-linear interactions between meteorological factors, terrain characteristics, and environmental risks. For example, previous studies show that gradient boosting and extreme value theory are more efficient than traditional boosting algorithms in predicting the amount of wildfires and burned area [7]. Meanwhile, there are studies that show that an ensemble model that combines Random Forest, XGBoost, and LightGBM has better performance on forecasting wildfires [14].

On the other hand, there are limitations in ensemble models, which is the trade-off required between computational and predictive performance. While processing large-scale data, the requirements for computational and hyperparameter optimization of training many learners are very concentrated. Besides, most of the literature focuses on an ensemble model and overlooks the comparison with other ensemble techniques. Therefore, it may bring about gaps in knowledge to choose the best application or method for us.

D. Neural Networks

Neural networks have been provided attention due to their ability to predict the environment and process non-linear relationships and large high-dimensional data. Among all types of neural networks, the Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and hybrid architectures have obtained significant effort on wildfire prediction, environmental risk evaluation, and pollution forecasting. CNN has been used in previous studies to predict the spread of wildfires since it may capture wind direction, vegetation density, and topography [15]. Moreover, deep reinforcement learning and transformer-based models have been mentioned in other studies on their applicability to predicting wildfire occurrences [10].

Although neural networks have various advantages, such as high accuracy of prediction and the ability to reveal hidden patterns in complex datasets, they have certain limitations, including high computation cost, high sensitivity to hyperparameters, and an unexplained black box nature. Previous studies point out that the design of neural networks should be cautious with steps of dimensionality reduction, and tuned to improve generalization and avoid overfitting [16]. However, neural networks can still be seen as an economic method in environmental data mining since they provide more complex risk prediction, early warning, and strategy support ability based on evidence.

E. Research Gaps and Contribution

Although there are improvements and developments in ensemble learning and neural networks in predicting

environmental issues, there are various limitations that exist. First and foremost, there is limited investigation on comparing various ensemble architectures, including Bagging, Random Forests, and Gradient Boosting, to ensure the best setting under different environments [12][14][15]. Moreover, there is limited comparison literature between ensemble models and hyperparameter-tuned neural networks. Therefore, there is uncertainty regarding which obtains the best performance on optimizing accuracy, interpretability, and computational efficiency in large data sets [16][17]. Furthermore, the existing literature may have insufficient structural validation methods, such as rigorous data partitioning and cross-validation. This may affect the reliability of evaluating the model performance and extrapolating to unseen environmental conditions [13][18].

Therefore, this paper fills the following gaps, which adopt various base learners and compare with a hyperparameter-optimized neural network. Besides, the systematic data division and model validation may help to ensure the reliability and replicability of the project outcome. Overall, this paper provides information based on evidence to predict the environmental risks and manage limited resources. It may help the decision-making process by providing practical and sustainable strategy support.

III. RESEARCH METHODOLOGY

This study implements a quantitative data mining approach using SAS Enterprise Miner to develop and evaluate predictive models for wildfire occurrences in California. The methodology focuses on the comparative analysis between Ensemble Models and Neural Network Models. The workflow is structured to address data quality issues and evaluate the relationship between model interpretability and predictive complexity under different data partitioning environments.

A. Problem & Data Understanding

This study uses the California Environmental Conditions dataset, which aggregates meteorological observations from the California Irrigation Management Information System (CIMIS) network. The dataset includes 128,010 environmental readings collected from 262 distinct weather stations that are geographically distributed across the state of California. The data spans from 2020 to 2025 and captures a five-year period characterised by significant climatic variability like drought phases and intense wildfire seasons. This longitudinal coverage is essential for training models that can generalise across different seasonal patterns and effectively predict the climatic precursors to wildfire events.

The dataset consists of 19 continuous environmental predictors and nominal identifiers:

TABLE I. DATASET VARIABLES TABLE

Feature	Type	Description
Target	Numeric	Binary indicator representing the occurrence of a wildfire.
Station ID (Stn_Id)	Numeric	A unique numeric identifier for each specific weather station

Station Name (Stn_Name)	String	The text name of the location where the station is situated
CIMIS Region	String	The broader geographic region in California where the station is located
Date	String	The specific date on which measurements were recorded.
Reference Evapotranspiration (ETo_(in))	Numeric	Measured in inches. Estimates the sum of evaporation from the soil and transpiration from plants.
Precipitation (Precip_(in))	Numeric	Measured in inches. The total amount of rainfall or other precipitation collected on that day.
Solar Radiation (Sol_Rad_(Ly/day))	Numeric	Measured in Langleys per day (Ly/day). This measures the intensity of the sun's energy hitting the earth's surface.
Average Vapor Pressure (Avg_Vap_Pres_(mBars))	Numeric	Measured in milliBars (mBars). Indicates the actual amount of moisture in the air.
Maximum Air Temperature (Max_Air_Temp_(F))	Numeric	The highest air temperature recorded during the 24-hour period, in Fahrenheit.
Minimum Air Temperature (Min_Air_Temp_(F))	Numeric	The lowest air temperature recorded during the 24-hour period, in Fahrenheit.
Average Air Temperature (Avg_Air_Temp_(F))	Numeric	The mean air temperature for the day, in Fahrenheit.
Maximum Relative Humidity (Max_Rel_Hum_(%))	Numeric	The highest percentage of humidity recorded during the day.
Minimum Relative Humidity (Min_Rel_Hum_(%))	Numeric	The lowest percentage of humidity recorded during the day.
Average Relative Humidity (Avg_Rel_Hum_(%))	Numeric	The average percentage of relative humidity for the 24-hour period.
Dew Point Temperature (Dew_Point_(F))	Numeric	Measured in Fahrenheit. The temperature to which air must be cooled to become saturated with water vapor.
Average Wind Speed (Avg_Wind_Speed_(mph))	Numeric	The average speed of the wind over the day, measured in miles per hour (mph).
Wind Run (Wind_Run_(miles))	Numeric	The total distance of wind that passed over the station in 24 hours (Average Wind Speed x 24 hours).
Average Soil Temperature (Avg_Soil_Temp_(F))	Numeric	The mean temperature of the soil, in Fahrenheit.

Prior to importing the data into the SAS modelling pipeline, a preliminary data cleaning process was conducted to address formatting inconsistencies and missing values. The raw dataset was initially converted to CSV format and column delimiters were standardised to ensure correct parsing. Data cleaning involves removing single quotes on categories such as CIMIS Region and Station Name to prevent categorical mismatching during the modelling phase. Missing values were identified and removed at categories such as Eto (83 rows), Average Soil Temperature (20 rows), Average Relative Humidity (8 rows), and Air Temperature (5 rows). The rows that contained missing values were excluded from the analysis to ensure that

all trained models relied on complete and valid environmental data.

B. Data Pre-Processing

To ensure the reliability of the predictive models, a pre-processing pipeline was constructed within the SAS Enterprise Miner program. This phase addressed three fundamental data quality challenges: the presence of sensor anomalies (outliers), the skewness of data, and the disparity in variable scales.

The environmental sensor networks are prone to measurement errors, which results in irregular readings that turn out to be extreme outliers. If the anomalies were left unaddressed, they can severely distort the decision boundaries of machine learning algorithms. To solve the issue, a Filter Node was added into the pipeline to eliminate inaccurate data. An upper threshold of 900ly/day was applied to the Solar Radiation (Sol_Rad) feature. Readings exceeding the threshold value were identified as sensor malfunctions and were excluded to prevent the models from learning invalid patterns. Furthermore, an upper limit of 1000 was established for the Wind Run feature as values beyond this threshold represent extremely irregular wind conditions for the region.. This filtering process ensures that the models are trained on valid meteorological data and aims to enhance the accuracy of the results that aligns with real-world conditions.

Statistical analysis of the input variables revealed that several features including Average Wind Speed, Precipitation, and Wind Run, exhibited right-skewness. Such skewness can negatively impact the performance of the models like Regression and Neural Networks. To address the issue, a Transform Variables node was utilised. A logarithmic transformation was applied to the features for a more normal distribution. This step aims to stabilise the variance across the range of the data and allows the regression algorithms to model relationships accurately and prevent exploding gradients in neural network models.

Other than that, a Standardisation technique was applied to all interval predictor variables using a second Transform Variables node. This process rescales the features to have a mean of 0 and a standard deviation of 1. This technique allows the high performance of the Neural Network and Gradient Boosting models. Without standardisation, the models would be biased toward variables with larger numerical scales and treat them more important due to their magnitude. By standardising the inputs, the study ensures that all environmental factors contribute equally to the model's learning process.

The target variable tells the occurrence of fire events: class 0 (non-fire events), class 1 (fire events). Due to the outnumber of non-fire events as compared to actual fire events, a Decision Node was implemented to balance the distribution. A custom weighting scheme was applied by assigning a significantly higher weight of 23.0 to the minority class compared to a weight of 1.0 for the majority class. This adjustment effectively counteracts the abundance of zero-values by penalising the model disproportionately for misclassifying a fire event.

C. Modelling Approaches

The modelling approach involves comparing two distinct architectures: Ensemble Models and Neural Networks. These models were evaluated under two data partitioning strategies to assess their robustness and sensitivity to training data volume.

Firstly the dataset was first partitioned into 60% Training, 30% Validation, and 10% Testing. This strategy was applied to Ensemble M1, Ensemble M2, and HP Neural M1 nodes. The allocation of 30% of the data to the Validation set allows for rigorous hyperparameter tuning and to ensure that overfitting does not happen during the iterative training process. The second partition was 70% Training, 20% Validation, and 10% Testing. This was applied to Ensemble M3 and HP Neural M2. This partition tests the hypothesis that complex, high-capacity models require a larger volume of training examples to effectively extract latent features and rare patterns from the data.

The primary modelling approach utilises an Ensemble architecture which aggregates the predictions of three distinct base learners including Decision Tree, Regression, and Gradient Boosting. The study contrasts a Baseline Ensemble (M1) against Optimised Ensembles (M2 & M3) which utilise manually tuned hyperparameters. For the optimised configurations, the Decision Tree components were tuned with a Maximum Depth of 10 and a Leaf Size of 20 to capture non-linear 'If-Then' relationships. Regression M2 and M3 nodes utilised Stepwise Selection to automatically filter redundant predictors and resolve multicollinearity among correlated variables. To address class imbalance, Gradient Boosting was configured with 100 iterations to allow the model to sequentially correct errors and improve the detection of rare fire events.

As a comparative benchmark, a High Performance (HP) Neural Network was trained as a comparative model. It is selected for its ability to model continuous meteorological interactions that may not be captured by binary tree splits. By comparing the Neural Network Model with the Ensemble Models, the study evaluates whether the predictive power of the neural network offers a significant advantage over the rule-based logic of the ensemble approach.

D. Evaluation Strategy

To validate the models, the data was divided into training, validation, and testing parts. The validation part was used during the modelling process to fine tune the parameters and select the best performing model to prevent overfitting. The Test part was then used to provide a final and unbiased assessment of how the model performs on completely unused data.

The choice of evaluation metrics was driven by the research objective of creating a reliable early wildfire warning system. The study prioritised Recall to determine how many actual fires were correctly detected, which is critical for public safety to avoid missed warnings. The F1-Score was used to ensure a balance between detecting fires and minimising false alarms, while the ROC Curve and Misclassification Rate provided an overall confirmation of the model's reliability and error rate.

IV. RESULTS AND DISCUSSION

This section discusses the results obtained from the classification models developed in this study, which are the ensemble model and the neural network model. The performance of these models will be evaluated based on metrics such as misclassification rate, accuracy, precision, recall, F1-Score and ROC curve. This section will also identify the significant environmental conditions that influence wildfire occurrence from the event classification table. Lastly, the discussion section will relate the findings from the models to sustainability decision-support goals and recommend wildfire mitigation strategies for the protection of California's critical forest infrastructure for future generations.

A. Model Comparison

The figure in appendix B, Figure 1 shows the fit statistics of the predictive models. The misclassification rate in the validation dataset is used to identify the top performing model between ensemble and neural network models. The results show that HP Neural M2 has the lowest misclassification rate of 0.032254 among the neural network models, whereas Ensemble M3 has the lowest misclassification rate among the ensemble models. However, HP Neural M2 performs better than Ensemble M3 in terms of misclassification. Both models use a data partition of 70% for training, 20% for validation and 10% for testing.

Besides, the figure in appendix B, Figure 2 shows the ROC chart, which compares the true positive rate to the false positive as the threshold for predicting '1' change for different models. The results show that HP Neural M2 achieved a most accurate test among the neural network models, whereas Ensemble M3 has achieved the most accurate test among the ensemble models. However, HP Neural M2 is more accurate as compared to Ensemble M3 based on the ROC Chart.

The figure in appendix B, Figure 3 shows the classification chart of Ensemble M3 and HP Neural M2 models for the validation set. Target 0 indicates no fire and Target 1 indicates fire events. The results indicate that HP Neural M2 chart shows a larger proportion of correctly identified fires as compared to Ensemble M3, which has a larger false negative.

TABLE II. EVENT CLASSIFICATION TABLE

Metric	Ensemble M3	HP Neural M2
True Positive (TP)	220	546
True Negative (TN)	24338	24117
False Positive (FP)	86	307
False Negative (FN)	841	515

Table I shows the event classification table for Ensemble M3 and HP Neural M2, which list out the values of true positive, true negative, false positive and false negative of the models. These figures are extracted and used to derive the accuracy, precision, recall, support and F1-score values shown in Table II.

TABLE III. SUMMARY OF THE PERFORMANCE EVALUATION

Machine Learning	Accuracy (%)	Precision (%)
Ensemble M3	96.36	71.9
HP Neural M2	96.77	64.01

Table II shows the summary of the performance evaluation of Ensemble M3 and HP Neural M2 in terms of accuracy, precision, recall and F1-score. Accuracy represents the ratio of correct predictions to the total number of observations, where correct predictions include both true positive and true negative outcomes. The results show that HP Neural M2 has achieved a higher accuracy of 96.77% as compared to Ensemble M3 of 96.36%, which indicates HP Neural M2 has achieved a better overall accuracy. Precision measures the accuracy of positive classifications and is known as the ratio of true positives to the sum of true positives and false positives. The results show that Ensemble M3 has a higher percentage of precision of 71.9% as compared to HP Neural M2 of 64.01%, which indicates that Ensemble M3 is highly effective at minimizing false alarms. Moreover, recall measures how effectively a model identifies positive cases by calculating the percentage of true positives that are correctly classified. The results show that HP Neural M2 has a higher percentage of recall of 51.46% as compared to Ensemble M3 of 20.74%, which indicates that HP Neural M2 has correctly identified more than half of all actual fire events in the validation dataset (Chung and Teo, 2023). The F1-score of HP Neural M2 (0.3219) is higher than that of Ensemble M3 (0.5705), which indicates that HP Neural M2 effectively minimizes both false positives and false negatives.

Based on the above evaluated metrics, it can be concluded that HP Neural M2 performs the best among other models. This is because it has the lowest misclassification rate and highest accuracy, recall and F1-score. The ROC chart also shows that HP Neural M2 is most accurate among the other models.

The figure in appendix B, Figure 4 indicates which factors most influence the prediction of wildfire occurrence in California. The results show that Stn_Name, which is the name of the CIMIS weather station is the most influential variable, followed by Sol_Rad which is the amount of solar energy received at the station. Mhawej et al. mentioned that solar radiation, which is also known as illumination intensity is a crucial factor that influences the occurrence of wildfires as it is the main driver of evapotranspiration [17]. Evapotranspiration is the combined processes of soil evaporation and plants transpiration. As the solar radiation increases, it accelerates the evapotranspiration, which leads to reduced moisture content in vegetation and soil. Hence, leading to drier vegetation that increases the chance of wildfire occurrence.

Besides, Avg_Soil_Temp, the average soil temperature is the third significant factor that influences the prediction of wildfire occurrence. Mhawej et al. mentioned that higher soil temperatures reduce the soil moisture, which promote the dehydration of ground fuels that increases the likelihood of a wildfire. Hence, lower soil temperature has higher moisture as it contains higher water content, which decreases the probability of occurrence of wildfire [17].

Furthermore, Max_Air_Temp, maximum air temperature is also one of the significant factors that influences the California wildfire occurrence. Platt et al. mentioned that higher air temperatures diminish the water content in forest vegetation and fuels, which leads to drying of fuels and soils required for ignition [18].

Moreover, Min_Rel_Hum, minimum relative humidity is also an influential factor in predicting the California wildfire occurrence. Tourma et al. mentioned that as relative humidity decreases, dry air draws the moisture out of the vegetation, which makes fuels like grasses highly flammable [19].

B. Discussion

This study has achieved the first objective, which is to evaluate and compare ensemble models with neural network models in predicting wildfire occurrence in California based on CIMIS weather stations. The results show that HP Neural M2 model is the most suitable model, which achieved an accuracy of 96.77% and a recall of 51.46%, which indicates that the model captured more than half of all the wildfire events. This model has also achieved a F1-score of 0.5705, which is reliable to be used as a decision-support system in predicting wildfire occurrence. The results align with the study by Shao et al., which mentioned that neural networks often outperform other models in prediction accuracy. However, the results of the study show that ensemble M3 demonstrated high precision. Although, it effectively minimizes false alarms, it achieved a low recall and high precision, indicating that the model is risky as it fails to identify 80% of actual fires.

Moreover, this study has also met the second objective which is to determine the most significant environmental predictors of wildfire risk in California. The results show that the variables station name, solar radiation, average soil temperature, maximum air temperature and minimum relative humidity are significant in predicting wildfire occurrence. This is supported by the study by Mhawej et al., which mentioned that solar radiation is critical in predicting wildfire occurrence, as high illumination intensity causes vegetation to dry out faster, decreasing the time required for fire ignition [17]. The variable average soil temperature is not consistent with the previous study by Buch et al. as a significant factor in wildfire prediction [20]. Minimum relative humidity is consistent with Dong et al., where it was mentioned that low relative humidity is the main factor in determining wildfire size and that wildfires often occur during winter, especially on dry days [21]. Moreover, maximum air temperature is consistent with Dong et al. where the study showed that 60.5% of the wildfires occurred when the temperature was higher than the climatological mean [21].

Furthermore, this study has also achieved the third objective which is to recommend wildfire mitigation strategies in California based on the predictive results. The results show that HP Neural M2 model is the most suitable model for predicting wildfire occurrence. Hence, relevant authorities can use the suggested model to issue red flag warnings and restrict activities in high-risk zones, where areas with low minimum relative humidity and high air temperatures are identified. Agencies can prioritize soil moisture mapping for targeted

areas identified by the model with the variable station name (Stn_Name) for UAV patrols instead of conducting broad monitoring, which optimizes resource allocation. As the HP Neural M2 model captures more fire events, it can be used to prepare firefighting crews and equipment in regions identified by the model where wildfire often occur before ignition, in order to reduce response time when wildfire occurs. Besides, the HP Neural M2 model can be utilized for prioritized vegetation management, especially during periods of high solar radiation and high average soil temperature. Authorities can schedule prescribed burns or mechanical thinning in areas identified by station name variable, where hotspots are detected before peak drying periods to reduce fine fuels [22].

This study has limitations which need to be addressed in future studies. This study has applied a weighting scheme during data preprocessing to solve the problem of outnumber of non-fire events and actual fire events. However, this may have influenced the model's sensitivity to wildfire signals. The results show that Ensemble M3 have achieved high precision but very low recall, as weighting scheme reduces false positive predictions and increases the likelihood of missing actual wildfire occurrences. Moreover, this study has also removed extreme values to ensure the model's stability, but this may eliminate important signals in detecting serious wildfire situations. Maggioros and Tsalkitzis discussed a situation where wildfires have strong non-linear relationships with meteorological variables, in which extreme meteorological values may significantly influence fire severity. Removing extreme values may obscure these complex patterns and weaken the model's ability to predict severe wildfire events [23]. The third limitation of this study is that it only focused on specific CIMIS weather stations data in California, which lacks generalizability. If the weather stations were installed in other places of California, the selected model might lack past information about wildfire occurrence for these locations, resulting in inaccurate predictions.

V. CONCLUSION

This study conducted a comparative analysis on ensemble models and neural network models to predict wildfire occurrences in California. The main objective of this study was to identify an accurate predictive model to support early warning systems. The results reveals that the HP Neural Network M2 outperformed the ensemble models and has achieved an overall accuracy of 96.77% and a recall of 51.46%. while the Ensemble M3 model demonstrated high precision of 71.9%, its low recall of 20.74% tells that the model is less suitable as it is prone to detection failures of actual fire events, which poses significant risks to human life and ecosystems. Furthermore, the study identified that Station Name, Solar Radiation, Average Soil Temperature, Maximum Air Temperature and Minimum Relative Humidity being the most significant environmental predictors of wildfire risks.

These findings support future decision making by providing a data driven framework for resource allocation. The performance of the HP Neural Network M2 model allows relevant authorities to issue accurate warnings and restrict activities in high-risk zones characterised by high solar radiation and low relative humidity. By utilising the identified

significant variables, stakeholders can transition to targeted interventions such as prioritising soil moisture mapping via UAV patrols and pre-positioning firefighting crews in relevant station locations that were identified as hotspots.

On the other hand, this study is subject to certain limitations. Firstly, the heavy weighting scheme used to address class imbalance at the Target variable likely contributed to conservative predictions and high false negative rate observed in the Ensemble M3 model. Secondly, the removal of extreme outliers during preprocessing may have filtered out valid signals associated with extreme meteorological conditions that contribute to severe wildfire events. Finally, the study was geographically limited to specific CIMIS weather stations, which may reduce how broadly the results can be applied. Future research should focus on alternative methods to handle imbalanced data, incorporating larger datasets with wider geographical coverage, and testing the framework in other environmental domains to enhance reliability and usefulness.

REFERENCES

- [1] C.Campen, "Wildfires and Climate Change", National Aeronautics and Space Administration. <https://science.nasa.gov/earth/explore/wildfires-and-climate-change/>, Accessed: Dec.29, 2025.
- [2] National Geographic Society, "Wildfires," National Geographic Education. <https://education.nationalgeographic.org/resource/wildfires/>, Accessed: Dec. 29, 2025.
- [3] California Department of Forestry and Fire Protection (CAL FIRE), "Statistics" CAL FIRE. <https://www.fire.ca.gov/our-impact/statistics#:~:text=INTERVAL-WILDLAND%20FIRES,the%20national%20Incident%20Management%20Situation>, Accessed: Dec. 29, 2025.
- [4] California Department of Forestry and Fire Protection (CAL FIRE), "Top 20 Most Destructive California Wildfires" CAL FIRE. <https://34c031f8-c9fd-4018-8c5a-4159cdf6b0d-cdn-endpoint.azureedge.net/-/media/calfire-website/our-impact/fire-statistics/top-20-destructive-ca-wildfires.pdf?rev=737a1073f76947b4a3bfb960b19f44c7&hash=7CA02D30D9BF46A32D5D98BD108BA26A>
- [5] A. Malik, M. R. Rao, N. Puppala, P. Koori, V. A. K. Thota, Q. Liu, J. Gao, "Data-Driven Wildfire Risk Prediction in Northern California," *Atmosphere*, vol. 12, no. 1, p. 109, 2021.
- [6] Y. Wang, X. Bi, Y. Cheng, Y. Du, L. Feng, and S. Wang, "The relationship between maternal environmental temperature exposure and preterm birth: A risk prediction based on machine learning," *Sustainable Cities and Society*, vol. 115, p. 105814, 2024, doi: 10.1016/j.scs.2024.105814.
- [7] J. Koh, "Gradient boosting with extreme-value theory for wildfire prediction," **Extremes**, vol. 26, no. 2, pp. 273–299, Jan. 2023, doi: 10.1007/s10687-022-00454-6.
- [8] R. Jiang and J. Zhang, "Leveraging AdaBoost and Bayesian Optimization to Enhance Decision Tree Models for Wildfire Prediction," in *Proc. IEEE Int. Geoscience and Remote Sensing Symp. (IGARSS)*, Chengdu, China, 2025, pp. 237–239, doi: 10.1109/IGARSS55030.2025.11243062.
- [9] E. D. Omar, H. Mat, A. Z. A. Karim, R. Sanaudi, F. H. Ibrahim, M. A. Omar, M. Z. H. Ismail, V. J. Jayaraj, and B. L. Goh, "Comparative analysis of logistic regression, gradient boosted trees, SVM, and random forest algorithms for prediction of acute kidney injury requiring dialysis after cardiac surgery," *Int. J. Nephrol. Renovasc. Dis.*, vol. 17, pp. 197–204, Jul. 2024, doi: 10.2147/IJNRD.S461028.
- [10] C. Vidal-Silva, R. Pizarro, M. Castillo-Soto, B. Ingram, C. de la Fuente, V. Duarte, C. Sangüesa, and A. Ibañez, "A comparative study of a deep reinforcement learning solution and alternative deep learning models for

- wildfire prediction,” *Appl. Sci.*, vol. 15, no. 7, art. no. 3990, Apr. 2025, doi: 10.3390/app15073990.
- [11] S. A. A. Abu El-Magd, A. Maged, and H. I. Farhat, “Hybrid-based Bayesian algorithm and hydrologic indices for flash flood vulnerability assessment in coastal regions: machine learning, risk prediction, and environmental impact,” *Environ. Sci. Pollut. Res.*, vol. 29, pp. 57345–57356, 2022, doi:10.1007/s11356-022-19903-7.
- [12] J. C. Bortty, P. K. Bhowmik, S. A. Reza, I. A. Liza, M. N. I. Miah, M. S. R. Chowdhury, and M. A. Amin, “Optimizing Lung Cancer Risk Prediction with Advanced Machine Learning Algorithms and Techniques,” *J. Med. Health Sci.*, vol. 5, no. 4, pp. 35–48, 2024. [Online]. Available: <https://al-kindipublisher.com/index.php/jmhs/article/view/8113/6863>
- [13] M. Wang, R. Zhang, B. Yan, C. Song, Y. Lv, and H. Zhao, “Prediction of Soil Pollution Risk Based on Machine Learning and SHAP Interpretable Models in the Nansi Lake, China,” **Toxics**, vol. 13, no. 4, p. 278, 2025. [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12030823/>
- [14] K. R. Keerthana, N. Mishra, and S. Naveen, “A Stacked Ensemble Model for Accurate Wildfire Prediction Using Meteorological Data and Bayesian Optimization,” in *Proc. 2025 Int. Conf. Emerging Technologies in Computing and Communication (ETCC)*, Bengaluru, India, 2025, pp. 1–8, doi: 10.1109/ETCC65847.2025.11108477.
- [15] M. Marjani and M. S. Mesgari, “The large-scale wildfire spread prediction using a multi-kernel convolutional neural network,” **ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci.**, vol. X-4/W1-2022, pp. 483–488, Jan. 2023, doi: 10.5194/isprs-annals-X-4-W1-2022-483-2023.
- [16] J. Shao, B. Li, W. Liu, and M. Chen, “Model building for regional ecological risk prediction and evaluation of prediction accuracy,” *Advances in Civil Engineering*, vol. 2021, Article ID 6209506, 8 pages, 2021. [Online]. Available: <https://doi.org/10.1155/2021/6209506>
- [17] M. Mhawej, G. Faour, and J. Adjizian-Gerard, “Wildfire likelihood’s elements: A literature review,” *Challenges*, vol. 6, no. 2, pp. 282–293, 2015.
- [18] W. J. Platt, S. L. Orzell, and M. G. Slocum, “Seasonality of fire weather strongly influences fire regimes in south Florida savanna-grassland landscapes,” *PLoS One*, vol. 10, no. 1, e0116952, 2015.
- [19] D. Touma, S. Stevenson, F. Lehner, and S. Coats, “Human-driven greenhouse gas and aerosol emissions cause distinct regional impacts on extreme fire weather,” *Nature Communications*, vol. 12, no. 1, p. 212, 2021.
- [20] J. Buch, A. P. Williams, C. S. Juang, W. D. Hansen, and P. Gentine, “*SMLFire1.0: A stochastic machine learning (SML) model for wildfire activity in the western United States*,” *Geoscientific Model Development*, vol. 16, no. 12, pp. 3407–3433, 2023.
- [21] L. Dong, L. R. Leung, Y. Qian, Y. Zou, F. Song, and X. Chen, “Meteorological environments associated with California wildfires and their potential roles in wildfire changes during 1984–2017,” *J. Geophys. Res.: Atmos.*, vol. 126, no. 5, e2020JD033180, 2021.
- [22] B. Cabiyo, J. S. Fried, B. M. Collins, W. Stewart, J. Wong, and D. L. Sanchez, “Innovative wood use can enable carbon-beneficial forest management in California,” *Proc. Natl. Acad. Sci. U.S.A.*, vol. 118, no. 49, e2019073118, 2021.
- [23] S. Maggioros and N. Tsalkitzis, “Wildfire danger prediction optimization with transfer learning,” arXiv:2403.12871, 2024.

APPENDIX A

OUTPUT FOR SCENARIO 1

a) Tool 1

TABLE IV. SUMMARY OF LITERATURE PAPERS

Reference	Data Type	Model Used	Key Findings
Wang <i>et al.</i> [6]	Meteorological temperature and environmental data	1. Logistic Regression 2. Random Forest 3. Gradient Boosting	Machine learning models have better performance compared to traditional statistical approaches due to their effectiveness in capturing non-linear environmental risk patterns.
Koh [7]	Meteorological and extreme wildfire data	Gradient Boosting with Extreme-Value Theory	The accuracy of forecasting the output of rare and extreme wildfire events has been enhanced by using a hybrid gradient boosting approach.
Jiang and Zhang [8]	Remote sensing and meteorological data	1. Decision Tree 2. AdaBoost 3. Bayesian Optimization	Optimised ensemble decision tree models achieved higher accuracy and stability compared to single-tree models.
Omar <i>et al.</i> [9]	High-dimensional numerical datasets	1. Logistic Regression 2. Gradient Boosted Trees 3. SVM 4. Random Forest	Ensemble models consistently outperformed linear models when handling complex and high-dimensional data.
Vidal-Silva <i>et al.</i> [10]	Meteorological wildfire occurrence data	1. Deep Reinforcement Learning 2. CNN 3. LSTM	Deep learning models achieved high predictive accuracy but required substantial computational resources and tuning.
Abu El-Magd <i>et al.</i> [11]	Hydrological and environmental indicator data	1. Bayesian Models 2. Hybrid Machine Learning Models	Hybrid machine learning approaches improved environmental risk prediction performance.
Bortty <i>et al.</i> [12]	Multivariate numerical risk factor data	1. Logistic Regression 2. Random Forest 3. Gradient Boosting	Ensemble learning techniques demonstrated stronger generalisation performance compared to single models.
Wang <i>et al.</i> [13]	Soil pollution and environmental monitoring data	1. Random Forest 2. XGBoost 3. SHAP-based Models	Tree-based ensemble models achieved high accuracy while maintaining interpretability through SHAP analysis.
Keerthana <i>et al.</i> [14]	Meteorological wildfire data	Stacked Ensemble (Random Forest, XGBoost, and LightGBM)	Stacked ensemble architectures outperformed individual learners by leveraging complementary model strengths.

Marjani and Mesgari [15]	Regional ecological risk indicator data	Convolutional Neural Networks (CNN)	CNN models effectively captured spatial wildfire patterns but required large datasets and high computational cost.
Shao <i>et al.</i> [16]	Regional ecological risk indicator data	1. Regression Models 2. Machine Learning Models	Machine learning approaches improved prediction accuracy over traditional regression techniques.

APPENDIX B

OUTPUT FOR SCENARIO 2

a) Tool 1

Fit Statistics								
Model Selection based on Valid: Average Profit (_VAPROP_)								
Selected Model	Model Node	Model Description	Valid: Average Profit	Train: Average Squared Error	Train: Misclassification Rate	Valid: Average Squared Error	Valid: Misclassification Rate	
Y	Ensmbl4	Ensemble M3	1.66675	0.025735	0.036065	0.026901	0.036374	
	Ensmbl2	Ensemble M2	1.65906	0.026494	0.036713	0.027632	0.036806	
	Ensmbl	Ensemble M1	1.63562	0.028130	0.038256	0.028914	0.038114	
	HPNNA2	HP Neural M2	.	.	0.029552	.	0.032254	
	HPNNA	HP Neural	.	.	0.029611	.	0.032987	

Fig. 1 Fit Statistics of Predictive models

b) Tool 2

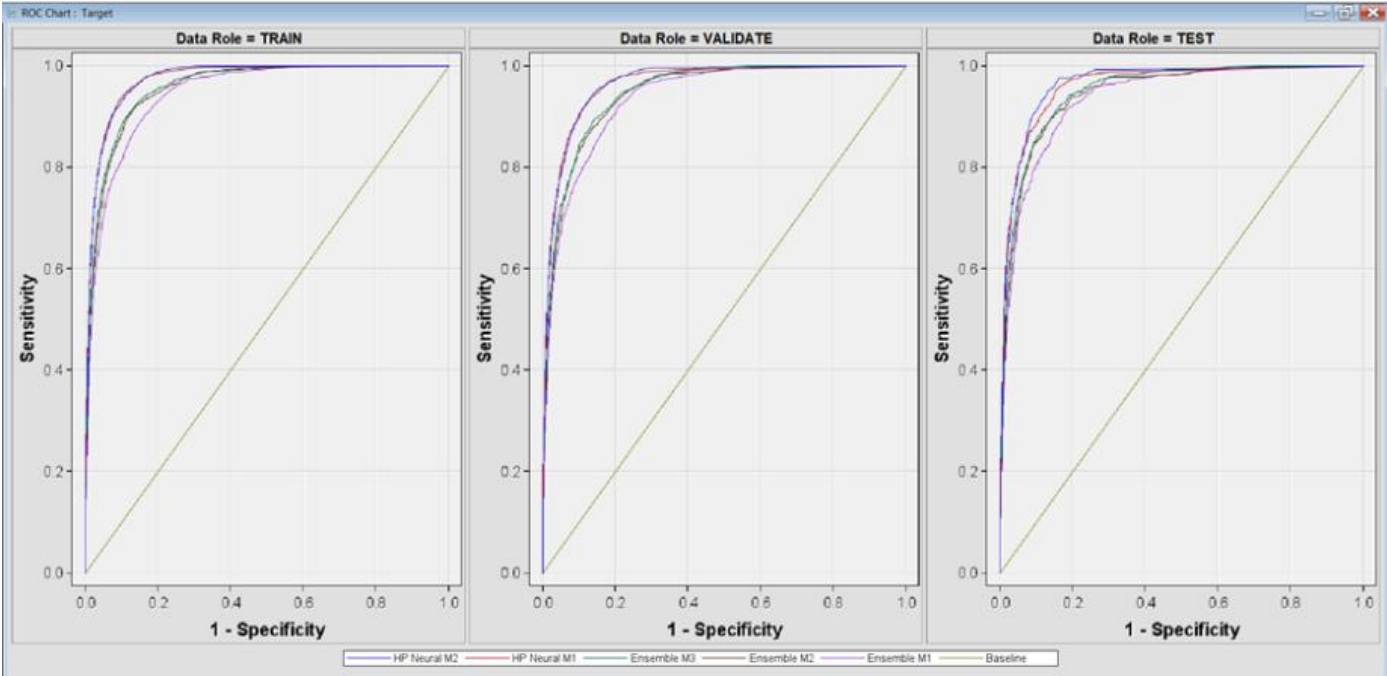


Fig. 2 ROC Chart

c) Tool 3

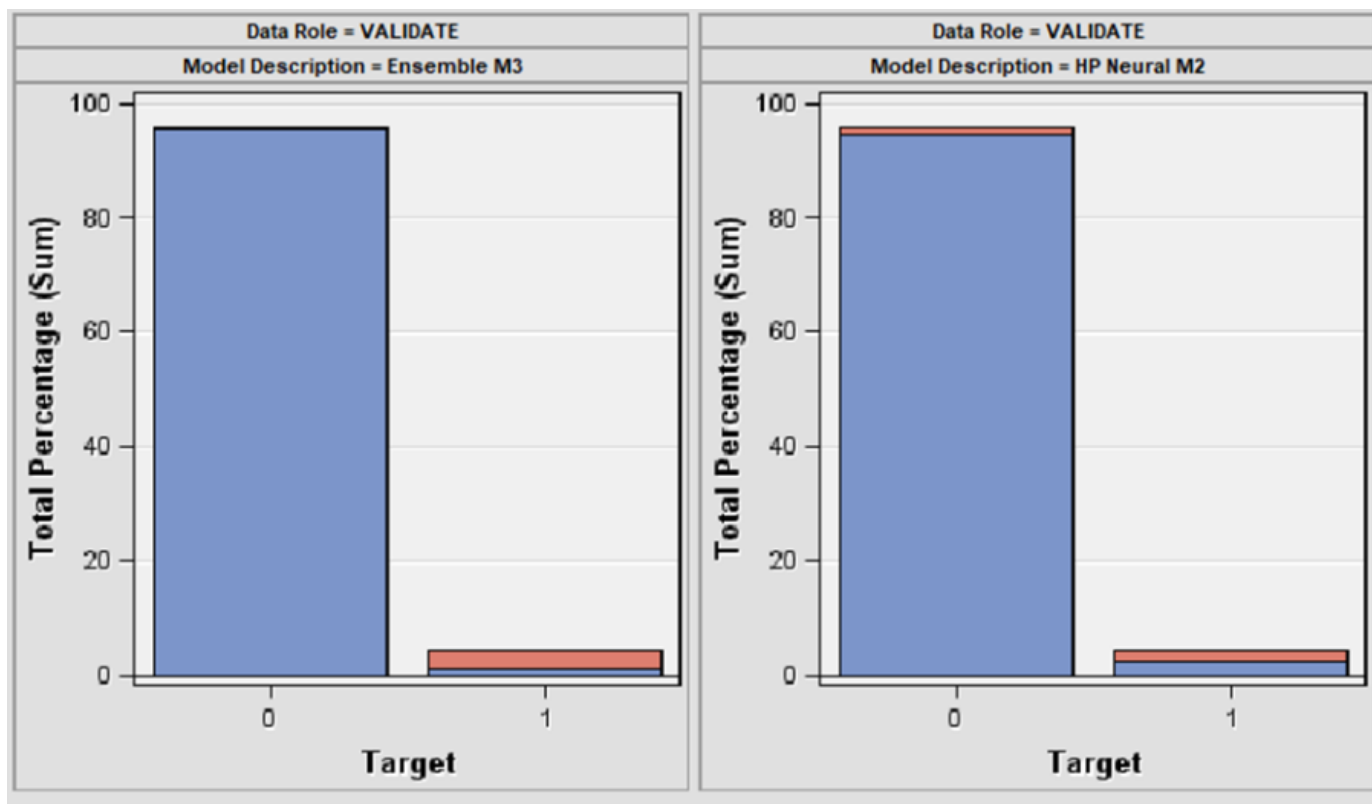


Fig. 3 Classification Chart of Ensemble M3 and HP Neural M2

c) Tool 3

Variable Importance

Obs	NAME	LABEL	NRULES	IMPORTANCE	VIMPORTANCE	RATIO
1	_Stn_Name		316	1.00000	1.00000	1.00000
2	STD_Sol_Rad	Transformed: Sol_Rad	86	0.80722	0.97794	1.21148
3	STD_Avg_Soil_Temp	Transformed: Avg_Soil_Temp	83	0.56000	0.69988	1.24979
4	STD_Max_Air_Temp	Transformed: Max_Air_Temp	47	0.43321	0.45196	1.04327
5	STD_Min_Rel_Hum_	Transformed: Min_Rel_Hum_	19	0.19619	0.19625	1.00033
6	STD_Dew_Point	Transformed: Dew_Point	28	0.18103	0.22957	1.26814
7	STD_Min_Air_Temp	Transformed: Min_Air_Temp	22	0.15156	0.13804	0.91077
8	STD_Avg_Vap_Pres	Transformed: Avg_Vap_Pres	12	0.14185	0.16547	1.16656
9	STD_LG10_Wind_Run	Transformed: Transformed: Wind_Run	26	0.12340	0.15885	1.28722
10	STD_Avg_Rel_Hum_	Transformed: Avg_Rel_Hum_	11	0.10154	0.09542	0.93978
11	STD_Avg_Air_Temp	Transformed: Avg_Air_Temp	11	0.08632	0.08130	0.94181
12	_CIMIS_Region		5	0.08105	0.02419	0.29838
13	STD_Max_Rel_Hum_	Transformed: Max_Rel_Hum_	6	0.07302	0.04381	0.59995
14	STD_Eto	Transformed: Eto	2	0.07077	0.08301	1.17293
15	STD_LG10_Avg_Wind_Speed	Transformed: Transformed: Avg_Wind_Speed	2	0.03065	0.03895	1.27049

Fig. 4 Variable Importance Ranking for Wildfire Predictors